

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.23 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 The area of region, in the first quadrant, bounded by the parabola $y = 9x^2$ and the line $x = 0$, $y = 1$ and $y = 4$, is
(A) $\frac{2}{9}$ (B) $\frac{14}{9}$ (C) $\frac{7}{9}$ (D) $\frac{11}{9}$
- Q.2 The area bounded by the curve $y = \ln x$ and the lines $y = 0$, $y = \ln 3$ and $x = 0$ is equal to
(A) 3 (B) $3\ln 3 - 2$ (C) 2 (D) $3\ln 3 + 2$
- Q.3 The area under the curve $y = |\cos x - \sin x|$, $0 \leq x \leq \frac{\pi}{2}$ and above x-axis is
(A) $2\sqrt{2}$ (B) $2\sqrt{2} - 2$ (C) 0 (D) $2\sqrt{2} + 2$
- Q.4 The equation of the curve passing through the origin and satisfying the differential equation $(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$, is
(A) $3(1 + x^2)y = 4x^3$ (B) $(1 + x^2)y = 3x^3$
(C) $3(1 + x^2)y = 2x^3$ (D) $(1 + x^2)y = x^3$
- Q.5 The area bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x-axis and lying in the first quadrant is
(A) 36 (B) 18 (C) $\frac{27}{4}$ (D) 9
- Q.6 The area of the region enclosed by the curves $y = x$, $x = e$, $y = \frac{1}{x}$ and the positive x-axis, is
(A) $\frac{e^2 - 3}{2}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{1}{2}$

- Q.7 The area bounded by the curve $y = \cos x$ and $y = \sin x$ between the ordinates $x = 0$ and $x = \frac{3\pi}{2}$, is
 (A) $4\sqrt{2} + 2$ (B) $4\sqrt{2} - 1$ (C) $4\sqrt{2} + 1$ (D) $4\sqrt{2} - 2$
- Q.8 The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$, is
 (A) $y = x + \ln x$ (B) $y = x \ln x + x^2$ (C) $y = x e^{x-1}$ (D) $y = x \ln x + x$
- Q.9 The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$, is equal to
 (A) $\frac{5}{3}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{4}{3}$
- Q.10 The area enclosed between the curve $y = \log_e(x + e)$ and the coordinate axes is
 (A) 1 (B) 2 (C) 3 (D) 4
- Q.11 The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom, then $S_1 : S_2 : S_3$ is
 (A) 1 : 2 : 1 (B) 1 : 2 : 3 (C) 2 : 1 : 2 (D) 1 : 1 : 1
- Q.12 The degree of the differential equation whose solution is $y = c_1 x + \sqrt{c_1}$, is
 (A) 2 (B) 3 (C) 1 (D) 0
- Q.13 If $y = c_1 e^{2x} + c_2 e^x + c_3 e^{-x}$ satisfies the differential equation $\frac{d^3 y}{dx^3} + a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$, then the value of $\frac{a^3 + b^3 + c^3}{abc}$ is equal to
 (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) $-\frac{1}{4}$ (D) $\frac{1}{4}$
- Q.14 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable functions satisfying $f''(x) = g''(x)$, $2f'(1) = g'(1) = 4$ and $3f(2) = g(2) = 9$. The value of $(f(4) - g(4))$ is equal to
 (A) -6 (B) -16 (C) -10 (D) -8

- Q.15 The value of positive real parameter 'a' such that the area of the region bounded by the parabolas $y = x - ax^2$, $ay = x^2$ attains its maximum value is equal to
 (A) $\frac{1}{2}$ (B) 2 (C) 1 (D) $\frac{5}{2}$
- Q.16 Number of solutions of the equation $\max(\sin \theta, \cos \theta) = \frac{1}{2}$ in $\theta \in (-2\pi, 5\pi)$ is equal to
 (A) 3 (B) 5 (C) 7 (D) 9
- Q.17 If $A(n)$ represents the area bounded by the curve $y = n/\ln x$, where $n \in \mathbb{N}$ and $n > 1$, the x-axis and the lines $x = 1$ and $x = e$, then the value of $A(n) + nA(n-1) =$
 (A) $\frac{n^2}{e+1}$ (B) $\frac{n^2}{e-1}$ (C) n^2 (D) en^2
- Q.18 Let $f(x) = ax^2 + bx + c$, where $a \in \mathbb{R}^+$ and $b^2 - 4ac < 0$. Area bounded by $y = f(x)$, x-axis and the lines $x = 0, x = 1$ is equal to
 (A) $\frac{1}{6}(3f(1) + f(-1) + 2f(0))$ (B) $\frac{1}{12}(5f(1) + f(-1) + 8f(0))$
 (C) $\frac{1}{6}(3f(1) - f(-1) + 2f(0))$ (D) $\frac{1}{12}(5f(1) - f(-1) + 8f(0))$
- Q.19 A point P moves in xy plane in such a way that $[x + y + 1] = [x]$ where $[x]$ is the greatest integer less than or equal to x, and $x \in (0, 2)$. Area of the region representing all possible positions of the point P is equal to
 (A) 2 (B) 8 (C) $\sqrt{2}$ (D) 4
- Q.20 Area enclosed by the curve which is defined parametrically as $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ (t is parameter) is equal to
 (A) π (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$
- Q.21 Solution of the differential equation $\cos x \, dy = y(\sin x - y)dx$, $0 < x < \frac{\pi}{2}$ is
 (A) $y \sec x = \tan x + c$ (B) $y \tan x = \sec x + c$
 (C) $\tan x = (\sec x + c)y$ (D) $\sec x = (\tan x + c)y$

Q.22 If $x \frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of equation is

- (A) $y \log \left(\frac{x}{y} \right) = cx$ (B) $x \log \left(\frac{y}{x} \right) = cy$ (C) $\log \left(\frac{y}{x} \right) = cx$ (D) $\log \left(\frac{x}{y} \right) = cy$

Q.23 The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1 and c_2 are arbitrary constants, is

- (A) $y' = y^2$ (B) $y'' = y' y$ (C) $y y'' = y'$ (D) $y y'' = (y')^2$

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ANSWER KEY

Q.1	B	Q.2	C	Q.3	B	Q.4	A	Q.5	D
Q.6	B	Q.7	D	Q.8	D	Q.9	D	Q.10	A
Q.11	D	Q.12	A	Q.13	C	Q.14	C	Q.15	C
Q.16	C	Q.17	C	Q.18	D	Q.19	A	Q.20	A
Q.21	D	Q.22	C	Q.23	D				

